Interval Data Structures

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Range Minimum Query (RMQ) Problem

- Given an array of numbers, you will be asked Q queries:
  L, R: Find min(a[L], a[L+1], …, a[R])
- If the array stays the same for each query, it is a static RMQ problem.
- Dynamic RMQ has updates as well, such as modifying any specific value of the array, or even range of values.

Static RMQ

- The array elements won’t change.
- Given an array of numbers, you will be asked Q queries:
  L, R: Find min(a[L], a[L+1], …, a[R])
- We already know how to solve it. How?

Online vs Offline solutions

- When you have to answer multiple questions, you are not bound to answer them one by one…
- Online solution: Read queries one by one, answer each immediately.
- Offline solution: Read all queries first, answer whole batch in the end.
- What is the advantage of knowing all queries beforehand?
- There are ways to prevent offline solutions for any problem.
  - Making the queries interactive. You only receive the next query after answering the previous one.
  - Encrypting the next query by the answer of the last query. So you can’t retrieve the query without answering last one.
- So how to do static RMQ in offline?

Static RMQ (Online)

- We use the “Sparse Table” data structure to solve this problem in,
  - O(N lg N) preprocess time and O(1) per query.

Sparse Table: Preprocess

```
const int N = 1e5+10;
const int L = 17;

int n;
int a[N];

int t[L][N];

void preprocess () {
    for (int i = 0; i < n; ++i)
        t[0][i] = a[i];
    for (int j = 1; j < L; ++j) {
        for (int i = 0; i + (1 << j) - 1 < n; ++i) {
            t[j][i] = min(t[j - 1][i], t[j - 1][i + (1 << (j - 1))]);
        }
    }
}
```

Sparse Table: Query

```
int rmq (int l, int r) {
    int k = __lg(r - l + 1);
    int res = min(t[k][l], t[k][r - (1 << k) + 1]);
    return res;
}
```

https://www.spoj.com/problems/POSTERIN/

All the buildings in the east district of Byteburg were built in accordance with the old architecture: they stand next to each other with no spacing in between. Together they form a very long chain of buildings of diverse height, extending from east to west.

The mayor of Byteburg, Byteasar, has decided to have the north face of the chain covered with posters. Byteasar ponders over the minimum number of posters sufficient to cover the whole north face. The posters have rectangular shape with vertical and horizontal sides. They cannot overlap, but may touch each other, i.e. have common points on the sides. Every poster has to entirely adjoin the walls of certain buildings and the whole surface of the north face has to be covered.

Task

Write a programme that:

- reads the description of buildings from the standard input,
- determines the minimum number of posters needed to entirely cover their north faces,
- writes out the outcome to the standard output.

Input

The first line of the standard input contains one integer N ($1 \le N \le 250000$), denoting the number of buildings the chain comprises of. Each of the following N lines contains two integers D_i and W_i ($1 \le D_i, W_i \le 10^9$), separated by a single space, denoting respectively the length and height of the i-th building in the row.

Output

The first and only line of the standard output should contain one integer, the minimum number of rectangular posters that suffice to cover the north faces of the buildings.

https://codeforces.com/contest/475/problem/D

Given a sequence of integers $a_1, ..., a_n$ and q queries $x_1, ..., x_q$ on it. For each query x_i you have to count the number of pairs (l, r) such that $1 \le l \le r \le n$ and $gcd(a_l, a_{l+1}, ..., a_r) = x_i$.

$\gcd(v_1, v_2, \ldots, v_n)$ is a greatest common divisor of $v_1, v_2, ..., v_n$, that is equal to a largest positive integer that divides all v_i.

Input

The first line of the input contains integer n, $(1 \le n \le 10^5)$, denoting the length of the sequence. The next line contains n space separated integers $a_1, ..., a_n$, $(1 \le a_i \le 10^9)$.

The third line of the input contains integer q, $(1 \le q \le 3 \times 10^5)$, denoting the number of queries. Then follows q lines, each contain an integer x_i, $(1 \le x_i \le 10^9)$.

Output

For each query print the result in a separate line.

https://codeforces.com/problemset/problem/514/D

An army of n droids is lined up in one row. Each droid is described by m integers $a_1, a_2, ..., a_m$, where a_i is the number of details of the i-th type in this droid's mechanism. R2-D2 wants to destroy the sequence of consecutive droids of maximum length. He has m weapons, the i-th weapon can affect all the droids in the army by destroying one detail of the i-th type (if the droid doesn't have details of this type, nothing happens to it).

A droid is considered to be destroyed when all of its details are destroyed. R2-D2 can make at most k shots. How many shots from the weapon of what type should R2-D2 make to destroy the sequence of consecutive droids of maximum length?

Input

The first line contains three integers n, m, k ($1 \le n \le 10^5$, $1 \le m \le 5$, $0 \le k \le 10^9$) — the number of droids, the number of detail types and the number of available shots, respectively.

Next n lines follow describing the droids. Each line contains m integers $a_1, a_2, ..., a_m$ ($0 \le a_i \le 10^8$), where a_i is the number of details of the i-th type for the respective robot.

Output

Print m space-separated integers, where the i-th number is the number of shots from the weapon of the i-th type that the robot should make to destroy the subsequence of consecutive droids of the maximum length.

If there are multiple optimal solutions, print any of them.

It is not necessary to make exactly k shots, the number of shots can be less.

Examples

input Copy

```
5 2 4
4 0
1 2
2 1
0 2
1 3
```

output Copy

```
2 2
```

What type of functions can we query?

- Idempotent function:

$$f(x, x) = x$$

- Examples: min/max, gcd, lcm, AND, OR.
- Can we not handle non-idempotent functions?

Point Update Range Sum problem

- Given an array of n integers, handle Q queries of type:

  1 i x: set a[i] = x
  2 l r: output a[l] + a[l+1] + … + a[r]

- We already know how to solve the problem without any updates using prefix sum technique. Now we will see how to handle updates.

Segment Tree

- A segment tree is a binary tree that contains the result of specific intervals of the array.
- Root node contains the result (in this case: sum of values) of all indices.
- If a node contains the information for the range [L, R] where L < R, then it has two children. Let mid = (L + R) / 2.
  Left child: [L, mid]
  Right child: [mid + 1, R]

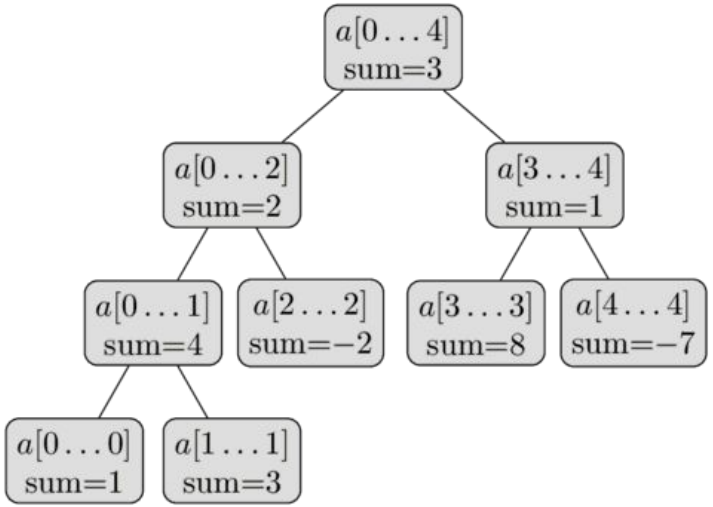


Image: cp-Algorithms

Building a Segment Tree

```
int a[N];
int t[MAX];

void build (int u, int b, int e) {
    if (b == e) {
        t[u] = a[b];
        return;
    }
    int mid = (b + e) / 2;
    build(2 * u, b, mid);
    build(2 * u + 1, mid + 1, e);

    t[u] = t[2 * u] + t[2 * u + 1];
}
```

Querying in a Segment Tree

```
int query (int u, int b, int e, int l, int r) {
    if (r < b || e < l) return 0;
    if (l <= b && e <= r) return t[u];

    int mid = (b + e) / 2;
    int lc = 2 * u, rc = 2 * u + 1;

    int left_contribution = query(lc, b, mid, l, r);
    int right_contribution = query(rc, mid + 1, e, l, r);

    return left_contribution + right_contribution;
}
```

Updating in a Segment Tree

```
void update (int u, int b, int e, int pos, int x) {
    if (b == e) {
        t[u] = x;
        //a[pos] = x;
        return;
    }
    int mid = (b + e) / 2;
    int lc = 2 * u, rc = 2 * u + 1;

    if (pos <= mid) update(lc, b, mid, pos, x);
    else update(rc, mid + 1, e, pos, x);

    t[u] = t[2 * u] + t[2 * u + 1];
}
```